1)

i: $Re =$ **-1/2** $Im =$ **5/3** i $Z =$ **-1/2+5/3** i **= -1/2-5/3** i

ii: $Re =$ **-3/2** $Im =$ **1** i $Z =$ **-3/2+1** i **= -3/2+1** i

iii: $Re =$ **0** $Im =$ **-12** i $Z =$ **0-12** i **= 0+12** i

iv: $Re =$ **4/5** $Im =$ **-5/4** i $Z =$ **4/5-5/4** i **= 4/5+5/4** i

2)

i: Z1+Z2-Z3 = (2+1i)+(-1+1i)-(2+3i) = 2+1i+(-1)+1i-2+3i = -1+5i

ii: Z1 Z2 Z3 = (2+1i)*(-1+1i)*(2+3i) = $(2*(-1) - 2 * 2 - (-1) * 2)$**+ i**$(1*1+1*3+1*3)$

= 0+7 i

iii: $\frac{2+1i}{(-1)+1i}$ **+** $z\frac{2}{1}$ **+** $z\frac{2}{3}$ $+$ $\frac{2+3i}{2+1i} =$ $\frac{2}{-1}+\frac{1i}{1i}+ z\frac{4}{4}+\frac{2}{2}+\frac{3i}{1i} = \frac{4}{1}+\frac{4i}{2i}+ z\frac{4}{4}$

3)

$Z^2 = 2\hat{z} - 1$

$2\hat{z} = \sqrt{X^2 + (-1)^2}$ **+**$\sqrt{X^2 - 1^2}$**=**

$2\hat{z} = X + 2$

4)

i) a: Modulus form: $|z1| = \sqrt{x^2 + iy^2}$ **argument: arctan**$\left|\frac{y}{x}\right|$

$\sqrt{1^2 + 1^2} = \sqrt{2}$ $\theta = \tan\left(\frac{1}{1}\right)$ $\tan\theta = 45$

ii: $|z2| = \sqrt{1^2 + (-1)^2} = \sqrt{1+1} = \sqrt{2}$ $\tan\theta = \frac{-1}{1}$ $\tan\theta = -45$

Polar formula: $z = (cos\theta + isin\theta)$

$z_1.z_2 = |z_1|(cos45 + isin45) * |z_2|(\cos(-45) + isin(-45)) = |z_1||z_2|(cos45 * \cos(-45))$ +

(isin45*isin(-45))+(isin45*cos(-45))-(isin(-45)*cos45)

b:

$i^{10} = -1$ $i^{20} = 1$ $i^{30} = -1$ $i^{40} = 1$ $i^{50} = -1$ $i^{60} = 1$ $i^{70} = -1$ $i^{80} = 1$ $i^{90} = -1$ $i^{100} = 1$